education sciences

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Systematic Review

Impact of Artificial Intelligence and Virtual Reality on Educational Inclusion: A Systematic Review of Technologies Supporting Students with Disabilities

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Abstract: The emergence of Artificial Intelligence (AI) and Virtual Reality (VR) technologies offers transformative potential for the advancement of inclusive education, particularly for students with disabilities. This systematic review critically evaluates the current state of research to assess the impact of AI and VR on enhancing educational accessibility, personalisation and social inclusion in education. AI-driven adaptive systems can dynamically tailor learning experiences to individual needs, while VR offers immersive, multi-sensory environments that promote experiential learning. Despite these advances, the review also identifies significant challenges, including the high cost of implementation, technical barriers and limited teacher readiness, which hinder widespread adoption. Ethical concerns such as privacy and algorithmic bias are cited as key areas that need careful consideration. The findings underscore the urgent need for further empirical research to explore the long-term impact of these technologies and advocate for more equitable access to AI and VR tools in underserved educational settings. Ultimately, the review highlights the importance of integrating AI and VR as part of a broader strategy to foster genuinely inclusive learning environments that align with the goals of the Convention on the Rights of Persons with Disabilities (CRPD).

Keywords: students with disabilities; artificial intelligence; virtual reality; systematic review; assistive technologies

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Citation: Chalkiadakis, A.; Seremetaki, A.; Kanellou, A.; Kallishi, M.; Morfopoulou, A.; Moraitaki, M.; Mastrokoukou, S. Impact of Artificial Intelligence and Virtual Reality on Educational Inclusion: A Systematic Review of Technologies Supporting Students with Disabilities. *Educ. Sci.* **2024**, *14*, 1223. https://doi.org/10.3390/educsci14111223

Academic Editor: Mike Joy

Received: 19 September 2024
Revised: 28 October 2024
Accepted: 5 November 2024
Published: 7 November 2024

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1. Introduction

The adoption of the Convention on the Rights of Persons with Disabilities (CRPD) by the United Nations General Assembly in 2006 marked a significant milestone in the global effort to recognise and protect the rights of individuals with disabilities [1].

According to the World Health Organization (WHO), disability is defined as a very broad term including but not limited to physical, cognitive, sensory or mental impairment leading to the substantial limitation of one or more major life activities [1,2]. Examples of physical disabilities (e.g., limited mobility), sensory disabilities (e.g., d/Deafness or blindness), intellectual disabilities (e.g., autism or Down's syndrome), and mental health problems (e.g., depression or anxiety) all fall under the umbrella term of disability. More than one billion people worldwide, or about 15% of the population, declare themselves to have some form of disability [1,2]. Among them, 1–2% have a physical disability,

5–6% are born with a sensory disability, and 1–3% have an intellectual disability. While neurological disorders, such as epilepsy, affect perhaps 1% of the world's population and mental disorders can affect up to 10–20% of people in their lifetime, advances in research into the neurobiology underlying disease are being translated into therapeutic applications at all levels and disorders of the nervous system [1,3].

Emphasis on inclusive education constituted the focal point of this convention, which mandates the elimination of barriers and equitable access to education for all students, regardless of their physical, cognitive, or sensory impairments through the transformation of educational systems [2]. The CRPD, alongside the International Classification of Functioning, Disability and Health (ICFDH) [3], advocates for a comprehensive rethinking of educational practices, requiring modifications in content, teaching methods, and institutional structures to accommodate the diverse needs of students with disabilities.

In this context, promising avenues for the enhancement of educational inclusion are appearing through the integration of emerging technologies such as Artificial Intelligence (AI) and Virtual Reality (VR). The utilisation of AI enables teachers and educators to tailor their educational materials and teaching methods accordingly, thanks to the AI tools' ability to recognise and analyse individual learning styles, needs, and preferences. Meanwhile, VR serves as a means for more interactive, experiential learning that can bridge the gap between theoretical knowledge and real-world application, thus fostering more open and inclusive educational settings that align with the objectives of the CRPD [4]. These technologies are not merely supplementary tools but rather essential components that can redefine instructional methods, enabling a more equitable classroom experience for all learners, regardless of their abilities or disabilities, as evidenced by recent advancements in adaptive learning frameworks that prioritise accessibility and inclusiveness in their design [5,6].

The potential of these technologies lies in their capacity to provide crucial elements for the support of the inclusion of students with disabilities in education, such as the personalisation of learning experiences, the engagement of students through immersive and interactive environments, and the adaptation to the unique needs of each individual. The importance of integrating these systems into traditional educational settings becomes more apparent in light of research, suggesting that inclusive education systems, which incorporate students with disabilities into general education classrooms, can effectively promote better academic outcomes for these students, as well as foster social skills and learning experiences for both them and their peers without disabilities, ultimately leading to more inclusive and accepting societies [7,8].

This systematic review seeks to evaluate the benefits, disadvantages and challenges of employing AI and VR technologies in educational contexts for students with disabilities. It examines the potential of these emerging technologies to facilitate personalised learning approaches and analyses existing research to determine effective practices as well as areas necessitating further investigation. Particularly, the review explores how AI and VR can be leveraged to align with the overarching objectives of inclusive education as outlined in the CRPD. This analysis will focus on the critical factors that influence the successful implementation of these technologies, including accessibility, content adaptability, and the need for inclusive representations within virtual environments, which are essential for creating equitable learning opportunities for all students, especially those with disabilities [9].

1.1. The 'Unseen' in Education

On the global level, access to education is not guaranteed for every individual. Educational opportunities and full participation in every aspect of social life are oftentimes not available for different demographic groups due to gender, religious, socio-economic and ethnic inequalities, as well as other exogenous factors, such as wars and disease [10]. However, disabilities demand a different approach, as they go beyond the scope of political, social or economic circumstances. They necessitate well-structured and comprehensive methods and strategies that will not only guarantee mere access to education, but also de-

velop an environment where people with disabilities are completely engaged and included, and able to optimise their skills and capabilities, irrespective of their disability.

A social model of disability is often claimed to be embraced by traditional educational systems; however, it is frequently observed that the specific learning needs and preferences, as well as the distinct identities of students with disabilities, are ignored, leading to a general sense of exclusion and detachment within the learning process [11–13]. Additionally, those students' diverse and unique nature of identities beyond their disabilities may be completely disregarded and they may even be categorised as passive "tragic victims" [14], who are supposed to uncritically comply with the various "interventions" and "treatments" suggested by professionals. This situation may result in the appearance of reductive and inaccurate identities for these students [15,16], causing them to avoid disclosing their disabilities due to fears of sidelining, misunderstanding or stigmatisation. Such misrepresentation not only directly affects their autonomy and agency, but also reinforces harmful prejudices and stereotypes that can seriously diminish and entirely impede their educational advancement and personal growth [7].

The Current Situation

There have been continuous efforts to develop regulations concerning the inclusion of people with disabilities in the educational environment, with CRPD [1] serving as the catalyst for the establishment of a framework that will provide effective and realistic approaches and solutions [17]. Even though governments have adopted measures and have developed strategies for fostering inclusive education, they have been proven inadequate to have a significant impact and to promote equal access and opportunities for each individual [18]. As a consequence, supranational organisations have shifted their attention to this issue, by actively engaging in discussions and publishing recommendations and frameworks aimed at promoting meaningful inclusion of students with disabilities, while respecting their unique cultural identities, such as those of the Deaf and Hard of Hearing (DHH) community. This approach seeks to balance integration with the preservation of distinct cultural identities. This has been broadly prevalent in the last decade, with declarations aiming specifically at this goal, based on the Sustainable Development Goals (SDGs) [19], such as the Incheon Declaration [20], which calls for the commitment to inclusive education and promotes an approach based on rights to quality and inclusive education.

Historically, the diverse needs of students with disabilities have been insufficiently addressed by educational institutions [21,22], which often are unable to provide students with equitable opportunities, inclusive experiences and necessary support within the classroom environment. This lack of suitable resources, adjustments and adaptations of the learning processes has fostered a reality under which students can choose between attending specialised programmes or integrating, to varying degrees, into mainstream programmes, thereby impeding their academic and social development [6]. This observation highlights the urgent need for transformative approaches that incorporate assistive technologies to create inclusive learning environments where all students can thrive, as shown by recent and ongoing discussions in the literature on the efficacy of adaptive learning tools in supporting students with learning difficulties within higher education [23].

The exclusion of students with disabilities from mainstream channels of education directly affects both them and their peers in various ways; the former do not have the capacity to receive the necessary tools and materials for their personal growth and academic achievement, while the latter is deprived of the opportunity to learn and grow in a diverse, inclusive and accessible environment, which is crucial for promoting the basis for the creation of a more empathetic, understanding, and accepting society [24]. As a result, for the development of an effective strategy towards inclusive education, a shift in both policy and practice is necessitated, emphasising collaboration among teachers and educators, administrators, and families to effectively address and overcome the consequences of the complex nature of teaching students with diverse needs and facilitate the successful integration of assistive technologies within the educational system [25,26].

This collaborative approach constitutes a crucial factor in addressing the barriers and hindrances faced by students with disabilities, as research [27–29] showcases that the utilisation of inclusive education practices and materials benefits these students and enhances the overall educational experience for everyone, reinforcing a sense of understanding and acceptance across all levels of education. The implementation of personalised support systems [30,31] and the continuous professional development of teachers and educators are required for a comprehensive transition towards the development of inclusive educational settings, which will foster the effective utilisation of assistive technologies [10,32,33], pedagogical strategies, and approaches [21] that accommodate the varied learning needs of students with disabilities, ultimately empowering these individuals to reach their full potential in academic and social spheres. These efforts are critical in creating an educational environment that values diversity, honours unique differences, and offers every student the opportunity to thrive and contribute to a more inclusive and equitable society [33]. Additionally, including these technologies in contemporary educational settings emphasises the importance of proactive support systems that address the unique learning challenges of students with disabilities, while also staying in line with the principles mentioned in the CRPD. For this to be achieved, there should be a thorough examination and understanding of the various and unique challenges faced by students, teachers, and educators in traditional educational environments [34], as well as of the existent need for training and resources [35] in order to provide teachers and educators with the knowledge and skills to effectively support and assist all students.

1.2. The Potential of AI and VR in Inclusive Education

With education being an ongoing topic of discussion and priority, it is observed that the introduction of AI and VR technology in education appears to have the capacity to offer a useful solution to the issues students with disabilities encounter [10,36]. These modern systems can be developed in specific ways to effectively address the issue of the provision of differentiated instructions in order to cater to each student's individual needs and subsequently enhance their engagement and interest. They can also provide specific learning approaches that may be friendlier to a number of students, compared to the traditional ones [4].

To be more specific, AI can be defined as the replication of human cognitive functions by machines, especially through computer systems [4,36]. This involves not only learning (acquiring information and rules for applying it), and reasoning (using rules to reach approximate or unambiguous conclusions) but also self-correction. AI is also used in education in the form of adaptive learning platforms that adjust content based on student performance, such as DreamBox, a math tutoring tool that constantly adapts and changes lessons based on the user's responses [4,10,36,37]. The use of AI in education opens up multiple simultaneous possibilities, such as personalization, real-time feedback, automation of repetitive tasks, analysis of learner needs, and the ability to adapt to the pace and needs of each learner. This makes AI particularly well suited for students with disabilities as it provides individualised learning experiences and tailored support.

Moreover, the utilisation of AI-driven adaptive learning tools can provide real-time feedback and tailored content that adjusts to each student's unique learning pace, enhancing their overall educational experience while simultaneously fostering a more engaging environment that promotes active participation among all students, including those with disabilities [37].

Additionally, students with disabilities are given the opportunity to work in a safe environment, where they acquire new knowledge and practise their skills through the utilisation of VR which is a technology that creates a simulated environment that users can interact with in a seemingly real way, using visual, auditory, and sometimes tactile feedback [37,38]. VR technologies provide immersive learning experiences and present real-world scenarios [38]. For example, VR is used in platforms like Labster, where students can conduct science experiments virtually, overcoming physical barriers to learning [39].

Immersive, experiential learning environments, simulation of real-world settings and the opportunity to practice in a risk-free environment are some of VR's affordances. Students with physical disabilities or who need to develop social skills, like students with autism, can practice interactions in a low-pressure virtual environment where these benefits are even more important. This serves as an important factor for bridging the gaps that currently exist between theory and practice [37,39]. Furthermore, these technologies can contribute to reducing anxiety and fostering the motivation of students, as they provide an appropriate controlled environment to develop their practical and social skills. This appears particularly beneficial for students who may have difficulties fitting into conventional classroom settings, such as those with autism or other learning difficulties [40].

These new emerging immersive technologies have the capability to address some physical, social, and cognitive barriers that have historically excluded students with disabilities from fully engaging and participating in the learning processes [6,37,41]. As a result, their integration within the existing educational system aligns well with the principles included in the CRPD concerning inclusive education. Furthermore, AI and VR systems not only play a crucial role in enhancing the academic achievements of students with disabilities, but they also foster a sense of belonging through the promotion of social interaction and collaboration among diverse student groups, which eventually leads to the creation of an inclusive atmosphere where understanding and mutual respect have a central place. It is crucial that appropriately developed technologies can benefit every student by creating an inclusive learning environment that enables students with different abilities and needs to learn together. AI and VR are resources that can support both students with and without disabilities by promoting inclusive engagement and collaborative learning opportunities.

This is particularly important considering recent advancements in virtual assistive technologies, which have given broad access to immersive educational experiences, paving the way for innovative learning opportunities that can transcend the limitations of physical classrooms and cater to a wide array of learning styles and needs [37,42,43].

In this regard, recent developments, such as the surge in investment from major technology companies to enhance the accessibility of virtual technologies, are presenting the shifting landscape of education. This enables institutions to adopt and possess tools that provide interactive, immersive and visually engaging teaching methods which are essential for fostering an inclusive environment where all learners can thrive. By developing universally accessible learning tools, AI and VR have the potential to support students across a broad range of abilities and foster an educational environment that emphasises equity, collaboration and inclusive participation for all learners [23,26,30].

2. Aims of the Study

This systematic review aims to critically analyse and synthesise existing research on the application of AI and VR technologies in educational settings for students with disabilities. The review seeks to explore the specific ways these technologies can contribute to enhancing educational inclusion, while also examining their associated benefits, drawbacks, and challenges as reported in the literature. The study is guided by the following research question:

> *How do AI and VR technologies contribute to enhancing educational inclusion for students with disabilities, and what are the associated benefits, limitations, and challenges in their implementation within educational settings?*

To address this question, the study is further guided by the following research objectives:

- To identify and assess the benefits of AI and VR technologies in supporting the educational experiences of students with disabilities;
- To investigate the disadvantages and limitations associated with the use of AI and VR technologies in educational environments for students with disabilities;
- To explore the challenges faced by teachers/educators and students in the implementation of AI and VR technologies within inclusive education;

- To identify the factors that contribute to the successful integration of AI and VR technologies in educational settings, particularly in supporting inclusive education.

3. Materials and Methods

To advance both theoretical and practical knowledge in the field, this study employed a systematic literature review approach, which was then complemented by a critical analysis approach in studying the papers under review. For systematic literature reviews, it is recommended that authors undertake a critical examination of the existing literature [44].

To be more specific, for this review, the recommendations of the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework [45] were followed. The protocol of the systematic review was registered on the Open Science Framework (https://doi.org/10.17605/OSF.IO/WA6XH, accessed on 22 April 2024 and updated on 18 October 2024) to ensure transparency and reproducibility of the review process.

3.1. Eligibility Criteria

The main criterion for inclusion in the review was that the studies were published in peer-reviewed journals indexed in Q1 and Q2 of SCOPUS database. This ensures that the selected research papers are of high quality and are recognised for their academic rigour and impact on the field [46].

Additionally, studies had to meet the following criteria: (a) focused on the use of AI, and VR technologies in relation to inclusive education for students with disabilities; (b) involved samples of students with disabilities across all educational levels (i.e., primary, secondary, higher education); (c) based on empirical quantitative, qualitative, or mixed-methods research designs; (d) written in English; and (e) published as of 2018 to reflect recent advances in AI and VR technologies in education [47].

Studies were excluded for the following reasons: (a) the absence of a focus on AI and VR technologies in relation to inclusive education for students with disabilities; (b) the exclusive focus on typically developing students; (c) theoretical and conceptual studies without primary empirical research; (d) the use of language other than English; (e) papers not published in peer-reviewed journals indexed in Q1 and Q2 of SCOPUS database (i.e., textbooks, discussion papers, and grey literature); and (g) studies published before 2018. These criteria (Table 1) were employed to refine the selection process and target the most pertinent literature for the review.

Table 1. Inclusion and exclusion criteria for SCOPUS-indexed articles Q1 and Q2.

Criterion	Inclusion	Exclusion
Period	2018–2024	Studies outside these dates
Languages	English	Studies in other languages
Type of article	Original research, published in a peer-reviewed journal	Articles that are not peer-reviewed or original research
Study design	Empirical studies (quantitative/qualitative/mixed methods)	Articles that lack empirical evidence (article/editorial/review)
Study focus/Population and sample	Students with a range of disabilities in all forms of education	Studies with typical developing students/without any disability
Literature focus	Articles that examined the impact of using Artificial Intelligence Technologies, such as AI and VR (Virtual Reality) in the educational context;	Articles that are not related to use AI and VR for educational purposes
	technologies such as AI and VR (Virtual Reality) in the educational context;	Articles that lack empirical evidence

Table 1. *Cont.*

Criterion	Inclusion	Exclusion
	studies comparing measures pre- and post-intervention	Studies that do not provide sufficient detail on intervention, population, outcomes and methodology for the overall analysis
	Studies with one or more of these findings: academic achievement, engagement and participation, accessibility of learning, personalisation of learning, social inclusion, barriers to technology integration and facilitators of successful implementation	

3.2. Information Sources and Search Query

The systematic search was conducted using the following databases: Education Source, ERIC, PsychINFO, Web of Science, and Scopus. The search strategy was tailored to each database, using the Boolean operators 'AND' and 'OR', to ensure comprehensive coverage and retrieval of relevant studies (Table 2).

Table 2. Sample of search terms for the ERIC database.

Step	Search Terms
1.	To identify relevant research through title and full text in order to generate a wide range of responses: artificial intelligen*(OR AI AND Virtual Realit* OR VR OR avatar* OR ChatGPT AND disabilit* OR disabled people OR disabled person OR physical disabilit* OR people with special need* OR handicapped OR autis* AND teacher* OR educator* AND student* with disabilit* OR learner* with disabilit* AND K12 OR school* AND elementary school* OR primary school* AND secondary school* OR high school* AND higher education OR tertiary education OR third level OR college OR campus OR postsecondary OR university OR tertiary)
2.	disabilit*(technolog* OR need* OR preference* AND challenge* OR barrier* OR obstacle* AND inclusive education AND integration OR inclusi* AND practice*)
3.	To identify relevant research through title and abstract in order to refine a range of responses: intervention (artificial intelligen* OR AI AND Virtual Realit* OR VR OR avatar* OR ChatGPT) AND disabilit*(OR disabled people OR disabled person OR physical disabilit* OR people with special need* OR handicapped OR autis* AND teacher* OR educator* AND student* with disabilit* OR learner* with disabilit*) AND K12 (OR school* AND elementary school* OR primary school* AND secondary school* OR high school* AND higher education OR tertiary education OR third level OR college OR campus OR postsecondary OR university OR tertiary) AND disabilit* AND technolog* OR need* OR preference* AND challenge* OR barrier* OR obstacle* AND inclusive education AND integration OR inclusi* AND practice*

The key search terms were the following:

1. artificial intelligence OR AI OR virtual reality OR VR OR avatar OR ChatGPT;
2. disability OR disabled people OR disabled person OR physical disability OR people with special needs OR handicapped OR autism;
3. teacher OR educator OR student with disability OR learner with disability;
4. K12 OR school OR elementary school OR primary school OR secondary school OR high school OR higher education OR tertiary education OR third level OR college OR campus OR postsecondary OR university;
5. 1 AND 2 AND 3 AND 4;

6. Limit #5 to 2018-current; English; peer-reviewed journal article.

The search took place from November 2023 to April 2024.

3.3. Study Selection Process

The study selection process is illustrated in Figure 1. Initially, 254 records were identified from databases, and 33 were found from registers. After removing 55 duplicates and 76 records marked as ineligible by automation tools, 156 records were screened based on the inclusion/exclusion criteria outlined in Section 3.2.

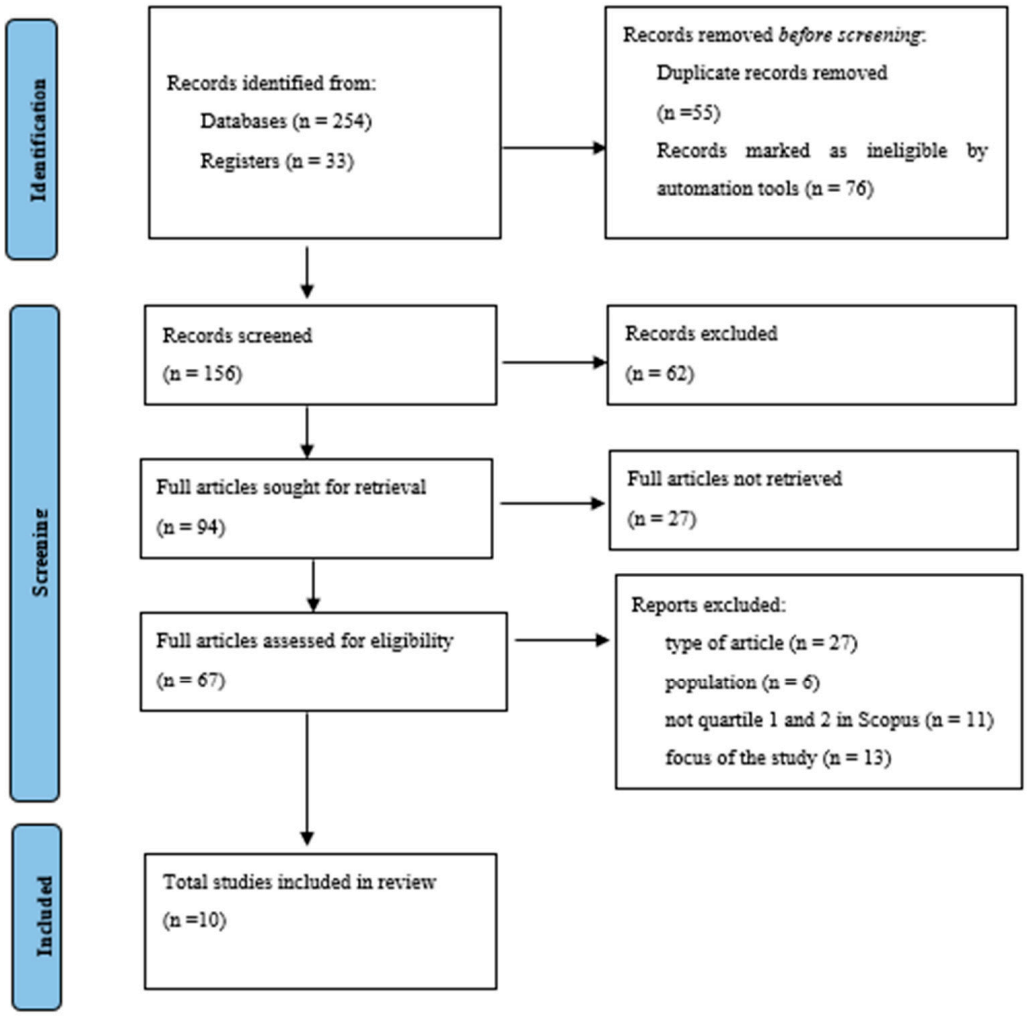

Figure 1. PRISMA flow diagram [48].

In the first round of screening, 94 full-text articles were sought for retrieval, of which 27 could not be retrieved. The remaining 67 articles were assessed for eligibility. Exclusion occurred for several reasons: 27 articles were excluded due to the type of article, 6 due to population focus, 11 because they were not published in journals indexed in Q1 and Q2 of the SCOPUS database, and 13 due to the study's focus not aligning with the criteria.

Data on the eligibility criteria were collected into an online spreadsheet that was accessible and editable by all authors of this review. After each of these searches, the research team met to screen the 'grey area' for example, papers that generally referred to AI and VR but did not necessarily relate directly to the incorporation of inclusiveness in education to students with disabilities. Finally, 10 of the articles were considered relevant and were selected for review in the ensuing categories.

More specifically, the 10 studies were selected from relevant academic journals that focus on educational technology, inclusive education and educational management. These journals include the *British Journal of Educational Technology*, *Journal of Special Education Technology*, *Disability and Rehabilitation: Assistive Technology*, *Education Sciences*, *ACM Transactions on Computer-Human Interaction*, *Journal of Information Systems Education*, *Journal of Interactive Media in Education and Computers and Education: Artificial Intelligence*. These top-tier publications ensured that the selected studies address the intersection of AI, VR and inclusive education for students with disabilities in different educational contexts.

3.4. Quality Assessment

To evaluate the quality of the studies, a checklist [49] consisting of ten questions was used to assess all the included articles based on three broad sections (Section A: "Are the results of the study valid?"; Section B: "What are the results?"; Section C: "Will the results help locally?"). The checklist acts as a pedagogical tool to determine the validity, consistency, methodological quality, presentation of results, and study outcomes. Each article was evaluated by one reviewer, and the results were subsequently reviewed by two additional researchers. Any discrepancies in the assessment were resolved through discussion to ensure reliability. The selected studies are marked with an * in the References section.

Each study was then subjected to a quality assessment based on this scoring process and grouped overall by quality. For every study a quality rating of high, medium or low, depending on how sound its methodology was according to recognised scientific standards and whether it answered any of the existing research questions. This categorisation allowed for an appropriate weighting, when analysing and summarising the results. Higher quality studies were weighted more heavily in the analysis and lower quality studies were interpreted with caution, particularly regarding their potential impact on the robustness of the overall conclusions.

4. Results

For the extraction and systemisation of the data useful for this systematic review, a standardised extraction form was used, and the components were reviewed and accepted by every member of the research team, in collaboration with three independent information search experts. Any discrepancies or disagreements were resolved through discussion between the reviewers, until consensus was reached. The information extracted from each article was as follows: author and year, country, research design, sample size, benefits, disadvantages, and challenges. The studies that were selected for this systematic review are presented in Table 3.

Table 3. Included studies from 2018 (in alphabetical order).

Ref. No	Author and Year	Country	Research Design	Sample Size	Results		
					Benefits	Disadvantages	Challenges
[50]	Atanga et al. (2020)	United States	Quantitative research (web-based survey)	62 schoolteachers	• Easier to use technologies • Bridging gaps between students	• Financial reasons hinder assistive technology use	• Inadequate teacher training
[51]	Coughlan et al. (2024)	United Kingdom	Qualitative research (online survey with open-ended questions)	138 university students with discloseddisabilities	• Improvement of knowledge about and practices for disability • Assistance in university tasks and courses • Enhancement of concentration • Help in daily organisation • Opportunity for audio and vision accessibility/multi-sensory learning	• Lack of systematic support	• Varied personal needs • Managing diverse support requirements
[52]	Gupta and Chen (2022)	United States	Qualitative research (interviews conducted through experimental chatbot Sammy with open-ended questions)	215 higher-education undergraduate students	• Supporting disadvantaged students • Provision of individual training • Same educational opportunities as other students • Accessibility due to simple and clear design • Instant feedback through interactive tools • Assistance with course material, socialising, and mental health issues • Inclusivity for non-native students		
[53]	Lister et al. (2021)	United Kingdom	Qualitative research (survey analysed with descriptivestatistics)	109 students with disabilities	• Ability to choose preferred means of communication and interaction • Flexibility and accessibility of AI technologies • Positive impact on students' wellbeing and empowerment • Increase in autonomy	• Misunderstandings due to machine's incapability to understand	

Table 3. *Cont.*

Ref. No	Author and Year	Country	Research Design	Sample Size	Results		
					Benefits	Disadvantages	Challenges
[54]	McGrath et al. (2023)	Sweden	Mixed methods research (survey with multiple choice and free-text questions)	194 higher education teachers	• Higher equity in outcomes • Creation of inclusive material	• Feelings of unfair treatment towards students without learning disabilities	• Teachers' fears and doubts about AI in higher education • Responsibility and fairness • Inadequate teacher training • Incorporation of AI in teaching processes
[55]	McMahon et al. (2020)	United States	Qualitative research (VR platforms: HTC VIVE VR goggles and Virzoom exercise bike (Virtual Reality gaming platforms). Data collection through the following: 1. smartwatches for the time of exercise, 2. Apple Watch for the heart rate and total calories burned.	School Students (Secondary) with intellectual and developmental disability (one student diagnosed with foetal alcohol syndrome, one student diagnosed with autism spectrum disorder, one student diagnosed with Down syndrome, one student diagnosed with intellectual disability and other health impairments)	• Increased students' physical activity • Increased exercise performance in terms of intensity and duration • Personalised approach tailored to the needs of each student.	• Expensive intervention	• Variability of individual responses or performance levels
[56]	Porayska-Pomska et al. (2018)	United Kingdom	Quantitative research (experimental methods using the ECHOES environment, with results collected through the SCERTS Assessment Process)	15 school students with Autism Spectrum Conditions (ASC) and 6 typically developing (TD) school students	• Increase in ASC students' responsiveness • More initiatives taken • Increase in autonomy	• ASC children show higher responsiveness only when they use AI technologies with a human practitioner • AI system's lack of responsiveness to changes	

Table 3. *Cont.*

Ref. No	Author and Year	Country	Research Design	Sample Size	Results		
					Benefits	Disadvantages	Challenges
[57]	Segbenya et al. (2023)	Ghana	Mixed methods research (questionnaire with closed and open-ended questions, using a four-point Likert scale and descriptive statistics)	294 postgraduate students	• Overcoming language and learning barriers	• Over-reliance on AI tools for university work • Lack of critical thinking	
[58]	Standen et al. (2020)	United Kingdom, Italy, and Spain	Quantitative research (data obtained through MaTHiSiS software)	67 school (secondary) students with disabilities (18 students with intellectual disabilities, 24 students with intellectual disabilities and some autistic tendencies, 18 students with Autism Spectrum Conditions, 7 typically developing students)	• Customization of activities • Identification of various affective states related to academic performance • Real-time response from the software	• Software's difficulty in distinguishing between different affective states	• Limited selection of educational material
[59]	Svensson et al. (2021)	Sweden	Quantitative research (data obtained from three test sessions, recording the impact of a six-week intervention that utilised a range of accessible assistive technologies on a battery of reading skills as well as student and parent reports regarding the apps' usefulness and functionality during the school day)	149 school (secondary) students with disabilities	Assistive Technology enhanced the following: • Reading skills; • Motivation and autonomy for school tasks; • Skills in the utilisation of applications.	• Difficulty in decoding and recognising words • Technical problems	• Difficulties in use

4.1. General Characteristics of Selected Studies

Studies from six different countries were included in this systematic review. Most of the studies were conducted in the United Kingdom (n = 4), followed by the United States (n = 3), and Sweden (n = 2). Ghana, Italy, and Spain are also included in the map, with one study being conducted in each. There is one transnational study which was conducted between the United Kingdom, Italy, and Spain. While the literature proves to be dispersed in terms of countries, there is a strong emphasis on studies conducted in Europe.

This systematic review contains studies with various research designs, equally present and included in the review, as shown in Figure 2. More specifically, out of the ten included studies, four follow the quantitative research model, four the qualitative research model, and two the mixed methods research design. This variety of study designs allows for a more comprehensive, thorough and multidimensional examination of the benefits, disadvantages and challenges that are linked with AI, VR and their utilisation in the educational processes for students with disabilities.

Figure 2. Number of papers per year based on research strategy.

In Table 4 the different types of disabilities included in the selected studies are presented. The classification of disabilities was based on the descriptions of disabilities included in CRPD [1]. Participants with physical (restricted mobility, manual dexterity issues) and sensory disabilities (deafness, blindness) were included in two studies [51,53]. Participants with autism were included in half of the selected studies [51,53,55,56,58], while other various intellectual disabilities [55,58] were also included. Foetal alcohol syndrome, a less popular disability, was also part of one of the selected studies [55]. Coughlan [51] and Lister [53] make mention of mental health-related disabilities, while also chronic health conditions, such as diabetes, asthma, and fatigue are included in their studies. Epilepsy, a neurological disability, is also included in the studies used for this systematic review [51,53]. Most of the selected studies (n = 7) mention communication disabilities [50–54,56,59], such as learning disabilities and speech impairment issues. Finally, there is one study [57] which includes participants with various disabilities in its research methodology, without, however, defining them.

Table 4. Type of disability per selected study according to CRPD descriptions of disabilities.

Selected Study	Type of Disability
[51,53]	Physical disabilities
[51,53]	Sensory disabilities
[51,53,55,56,58]	Intellectual disabilities
[51,53]	Mental health disabilities
[51,53]	Chronic health conditions
[51,53]	Neurological disabilities
[50–54,56,59]	Communication disabilities
[57]	Various/Non–defined disabilities

4.2. Benefits of AI and VR Technologies

Since the introduction of AI and VR technologies in the educational context, the literature has shifted its attention towards the positive impact that these can have on the educational progress and development of students with disabilities. Every study selected for this review included at least one element that proves to be beneficial, making it thus imperative to systemise and categorise those benefits that are more than prevalent in the literature.

One result that came up significantly/systematically throughout the examination of the literature was the AI technology's accessible and easy-to-use design [50–54] which enables every student, no matter their distinctive educational needs and preferences, to easily and effectively use these technologies and access valuable material, providing thus multisensory learning opportunities to everyone [51]. Additionally, the results from empirical studies have shown that, in fact, higher academic performance can be achieved, when AI and VR technologies are effectively incorporated in the educational processes [50–52,54,59]. This is either achieved through the increase in user-friendly, interactive platforms [51] or through the increased development of students' educational skills, such as reading or writing [59].

Furthermore, AI and VR technologies assist in the integration of students with disabilities in the educational processes, through the customised and individual training they are able to provide [51,52,55,58], covering the educational gaps that exist due to the inflexibility of the traditional materials. They also play a significant role in empowering and incentivising students with disabilities to take initiative, develop their autonomy, and enhance their mental health [52,53,56,59], as well as their social skills [52].

Another way through which AI technology is assisting students with disabilities is its well-developed response systems, which can give immediate and personalised feedback on students' performance [52,56,58], helping them fill gaps and organise their work [51]. Even further, AI systems can be used for the identification of students' affective states related to academic performance [58].

Apart from academic development, VR technology provides students with great amounts of physical exercise, connecting thus the learning process with students' personal well-being [55]. Non-native or students with language difficulties can also take advantage of the new technologies' functions which make the materials available in various languages or presented in a more simplified way [52,57]. Finally, digital skills acquisition can also be assisted and supported through the use of these newly developed technologies [59].

4.3. Disadvantages in Educational Environments

Apart from the benefits showcased above, the literature has tried to examine and shed light on the disadvantages that this sudden eruption of AI and VR technologies has brought in educational environments, elements that are clearly showcased and highlighted in the results of the empirical studies used for this systematic review.

One important drawback of the use of AI and VR technologies in educational processes is their inability to understand and process every kind of information they are receiving, leading sometimes to misleading or confusing responses from the system [53,56,58,59]. In

connection with this, the various different technical problems that come up during their utilisation shall be added [59].

Additionally, the cost to acquire some of these technologies constitutes a deterrent factor that blocks access to this type of multi-sensory materials [50,55]. This is why it is difficult for a widespread systematic supporting approach to be implemented [51], as not every student possesses the same resources to gain access to these assistive technologies tailored to their needs. Segbenya et al. [57] in their research dealt with the issue of over-reliance on these assistive technologies, showcasing the students' lack of critical thinking to process and manage new information, due to the simplicity of accessing and reaching huge numbers of information in a short period of time.

Even though AI and VR technologies, in general, seem to improve the academic and educational performance of students with disabilities, studies have shown that their effectiveness increased when their use was complemented with the assistance of a human practitioner [56], causing troubles to cases when they were autonomously used with no further guidance, which may end up being confusing for the students. Finally, there have been instances of students without any learning disabilities that state that the use of assistive technology constitutes an unfair advantage for students with disabilities and hinders the educational process [54].

4.4. Challenges in Inclusive Education

Another significant focus point of the literature is on how the integration of new assistive technologies in the educational context is followed by many challenges that both teachers and students have to deal with. There seems to exist a significant difficulty in efficiently incorporating AI and VR technologies in teaching processes [54], which can be explained both through the inadequate training teachers have received when it comes to the utilisation of new technologies [50,54], as well as through the teachers' own doubts and fears surrounding the use of assistive technologies in their classes, and the subsequent impact they would have in their work [54].

On the other hand, Svensson et al. [59] put emphasis on the complex nature of certain assistive technologies which are considered too inaccessible or elaborated for students with disabilities to navigate through them. The limited educational materials available in AI environments also present a significant challenge, according to Standen et al. [58]. Furthermore, AI and VR technologies must be adapted in order to satisfy a large number of varied personal and educational needs of students [51,55], making it almost impossible for one-size-fits-all solutions to be found. Finally, the literature also focuses on the issue of promoting ethical and responsible methods of integrating these assistive tools, so that they can effectively and appropriately complement every part of the learning experiences for the students [54].

5. Discussion

5.1. Integration of AI and VR Technologies in Inclusive Education: Synthesis and Comparative Analysis

The powerful transformative potential of AI and VR technologies could offer notable benefits for students with disabilities, once integrated into the educational framework [60,61]. Following the recent widespread development of assistive materials, these innovative technologies are progressively recognised as indispensable parts for the enhancement of student engagement, the provision of more accessible educational settings, and the facilitation of personalised learning. The findings from the 10 studies included in this systematic review, alongside recent research, highlight the methods and strategies in which AI and VR technologies are redefining the educational environment for students with disabilities, while also bringing to the forefront significant limitations and challenges that currently exist.

As emphasised in the literature [50,54], platforms developed with the assistance of AI have enabled teachers and educators to tailor their content and educational material to

the specific needs and preferences of students with disabilities, promoting the fostering of personalised instruction. While teachers and educators are certainly capable of adapting instruction on their own, the unique contribution of AI lies in its ability to improve both the efficiency and scalability of these efforts. AI-powered systems significantly reduce the cognitive and administrative overhead associated with personalising all aspects of instruction for different learners. This allows teachers and instructors to focus more on pedagogical interventions while ensuring that content is personalised to individual student needs in real time. In this way, AI facilitates more sustainable adaptive teaching practices, particularly in classrooms where addressing the varied requirements of multiple students simultaneously would otherwise be a considerable challenge.

This argument is further supported by the work of Hocine and Sehaba [62], who focused their attention on showcasing the assistive technologies' capacity to cultivate innovative and adaptive learning settings that can decisively transform the educational experiences of students, through their real-time data processing features which allow for the adjustment of material based on the needs of each individual student. The objective of personalised education aligns with the principles of Universal Design Learning (UDL) [63] which endorse a variety of means of expression, representation, and engagement. Nevertheless, the AI's inability to handle more complex, context-dependent interactions constitutes a significant limitation to the effective transformation of educational environments, as Fenu et al. [64] caution. In their research, Bhatti et al. [65] also make mention of this concern, highlighting the AI's struggle to process and assess more nuanced interactions, specifically in the context of special education. For example, Table 5 provides a detailed overview of how AI supports different disabilities, such as providing tailored support for complex tasks for students with neurological disabilities, but also highlights limitations, such as the difficulty of AI to account for the variability of these conditions [51,53]. While VR is of great benefit to students with intellectual disabilities as it promotes social interaction, its high cost is a major obstacle [55].

Table 5. AI and VR support and limitations for each disability type.

Disability Type	AI Technology Support	VR Technology Support	Limitations of AI	Limitations of VR	Ref. No
Physical Disabilities	Voice commands, and screen readers for accessibility.	N/A	Limited in addressing specific mobility needs.	N/A	[51,53]
Sensory Disabilities	Text-to-speech for visual impairments, and captioning for hearing impairments.	N/A	Struggles with mixed sensory impairments.	N/A	[51,53]
Intellectual Disabilities	Personalised learning content, and real-time feedback.	VR helps increase physical activity and social interactions.	Difficulty recognising subtle emotional states.	High cost and complexity.	[51,53,55,56,58]
Mental Health Disabilities	Chatbots provide emotional support.	N/A	May not detect nuanced emotional states.	N/A	[51,53]
Chronic Health Conditions	Adapt learning materials for pacing and fatigue.	N/A	Lack of individualised adaptations for specific conditions.	N/A	[51,53]
Neurological Disabilities	Tailored assistance for complex tasks.	N/A	Difficulty addressing variability in neurological disabilities.	N/A	[51,53]
Communication Disabilities	Chatbots and text-based tools for speech impairments.	N/A	Over-reliance can limit independent skills.	N/A	[50–54,56–59]
Various/Non-Defined Disabilities	General support via adaptable learning pathways.	N/A	Challenges adapting to diverse disability needs.	N/A	[57]

VR technologies have been proven to constitute essential components and tools for the engagement of students with intellectual and sensory disabilities, as presented in studies by Sánchez et al. [10] and McMahon et al. [55]. The multisensory, interactive environments that are offered by these immersive VR systems can be beneficial for students who have difficulties integrating into traditional educational settings. Zhang et al. [66] in their recent research have supported this idea, presenting how students with autism can develop and practise their social and problem-solving skills through simulations of real-world scenarios that VR technologies are offering. Furthermore, certain VR technologies can be technically complex to navigate through, posing a potential challenge for students with disabilities who may need additional support in order to use them effectively.

Recent literature has tried to explore the capacity of AI-driven assistive technologies to foster social inclusion for students with disabilities [5,67]. It is argued that interactive environments provided by AI systems can facilitate collaboration and enable constructive social interactions between peers, thus enhancing social integration and reducing social isolation. AI-powered tools such as virtual agents, chatbots and collaborative platforms enable peers to participate in teamwork by encouraging equal participation and supporting personalised interactions through adaptive feedback, creating a more inclusive learning atmosphere. However, AI should be carefully implemented and continuously monitored in order to prevent the appearance and reinforcement of biases and social hierarchies, especially against marginalised student groups, as Zajko [68] highlights. Furthermore, while AI may increase engagement and reduce boredom among learners with intellectual disabilities, it has not been proven to make a significant difference in achievement [58].

An additional issue that the literature has shifted its attention to is the financial costs that are linked with the implementation of AI and VR technologies in education. Atanga et al. [50] and Lim [22] showcase the constraints that are being imposed by the high costs of these systems, restricting their accessibility to more affluent schools and increasing gaps in the educational experiences of students. As McMahon et al. [55] and Kavanagh et al. [69] have argued, their widespread utilisation is being hindered by their high costs. This constitutes a significant deterrent factor for their integration in educational settings, especially in low-resource ones. Srinivasa et al. [70], having a similar approach, argue that for under-resourced educational institutions to access this type of technology, public–private partnerships or a substantial investment should be made.

Additionally, research has pinpointed some ethical concerns, including issues of potential for algorithmic bias and data privacy. Brewer [7] and Knox et al. [61] acknowledge the substantial educational benefits that AI and VR technologies provide; however, they advocate for the inclusion of safeguards against misuse and bias, to mitigate inequalities and reinforce a more equitable learning environment.

Despite the aforementioned challenges, all selected studies focus on the significant power of AI and VR in supporting diverse learners and promoting inclusive educational environments. For instance, Gupta and Chen [52] showcased the catalytic role of AI-based chatbots in the provision of personalised training and support for students with mental health and social difficulties, including non-native speakers. On a similar note, Porayska-Pomsta et al. [56] demonstrated the skills that students with Autism Spectrum Conditions acquired, including greater autonomy and responsiveness, when a hybrid approach blending AI and VR technologies with the work of human practitioners, concluding that it constitutes the most effective teaching model.

On the other hand, Coughlan et al. [51] highlight the importance of personalised and detailed information from students to ensure AI and VR technologies can truly prove inclusive and effective rather than promoting their direct benefits.

The heterogeneous nature of the findings suggests that assistive technologies do not universally apply across every type of disability or educational context. The availability of technical, human, and financial resources keeps playing an important role in the effectiveness of these systems, notwithstanding their notable benefits in terms of accessibility, personalisation and engagement. Additionally, Segbenya et al. [57] caution that the risk of

over-reliance on these tools is prevalent and could potentially hinder the development of critical thinking skills, especially in cases where students are totally dependent on them for the completion of their tasks. This serves as a broader area of concern in the literature about the realistic potential of AI to diminish creativity and reduce human agency in the learning process.

5.2. Appropriateness of AI and VR for Students with Disabilities

Many different factors, such as the specific needs of the students, the educational setting, and the availability of support structures within educational institutions, should be taken into consideration when assessing the suitability of introducing AI and VR technologies in the learning experiences of students with disabilities. The selected studies analysed in this systematic review highlight the potential of these systems to address the diverse needs of students with sensory, physical, cognitive, and social/emotional disabilities, even though their effectiveness is highly contingent on the methods and strategies employed for their implementation and integration in the educational environment.

AI has demonstrated its ability to serve as a great means of support for students with cognitive disabilities by allowing teachers and educators to tailor their material to the learners' cognitive level. AI-based platforms have been shown to reinforce and promote comprehension and retention by dynamically adjusting the complexity of educational materials, as supported by existing research [6,71]. Similarly, AI-driven chatbots have been demonstrated to offer personalised support, catering to the needs of students with learning and cognitive difficulties, and improving their learning experience and outcomes [52]. Furthermore, Lister et al. [53] prioritise the long-term equity of AI to ensure that the specific needs and concerns of students are met. They emphasise the importance of involving disabled students to achieve the creation of accessible and effective AI technologies. Nevertheless, it is crucial to note that the effectiveness of AI is directly linked to the extent to which teachers and educators can adequately and successfully connect these technologies to more traditional teaching approaches. The quality of teacher and educator training also plays an important role in this. When these two factors are absent, it becomes impossible for AI to reach its full potential as a supportive tool for students with cognitive and learning disabilities.

VR technologies provide students with immersive, multisensory tools that assist the learning experience of students and enhance their comprehension and engagement. As demonstrated by the studies of Sánchez et al. [10] and Svensson et al. [59], VR technologies can enhance the accessibility of abstract, complex concepts, due to their capacity to simulate real-world scenarios and to provide students with hands-on experiences. Furthermore, students with sensory impairments can drastically be assisted by VR, through the improvement of their spatial awareness and problem-solving skills [52]. However, due to the complex nature of certain VR environments, when it comes to students with severe sensory impairments, the integration of additional assistive systems is necessitated to ensure full accessibility, which may pose an additional challenge to the implementation of these tools. It is therefore important for new, more inclusive and adaptive VR technologies to be developed to tailor to the diverse needs of students.

Students with physical disabilities can also benefit from AI-driven adaptive systems and VR platforms, which give the opportunity to mitigate any existing physical barriers to educational experiences. As demonstrated by Page et al. [21], alternative input methods, such as eye-tracking and voice commands, can effectively integrate students with physical impairments in the learning process and enhance their engagement. Similarly, McMahon et al. [55] present the methods in which VR platforms can facilitate the engagement of students with developmental and physical disabilities in physical activities, thereby fostering their inclusion and participation.

6. Limitations

This review acknowledged several limitations. It focused solely on peer-reviewed journal articles, excluding potentially valuable grey literature and non-English studies. This may have overlooked important research from diverse cultural contexts. Cultural and linguistic differences in conceptualising and addressing disabilities in education could offer valuable insights not captured here. More research is deemed necessary to bring further insights from an international perspective in order for the research model to be further developed, to include other nations and to obtain better validation and more widely applicable conclusions. The extension of the studies to low- and middle-income countries would provide significant insights into how AI and VR technologies work in different educational contexts and would contribute to a more comprehensive understanding of their global applicability.

To gain a more comprehensive understanding of the global impact of AI and VR, future research should expand the scope to incorporate non-English studies and grey literature.

The diversity of included studies, in terms of research design, geography, and disability types, poses challenges for drawing generalised conclusions. Most studies were conducted in high-income countries where resources facilitate advanced technology use in education. This limits the applicability to low- and middle-income educational contexts, which may differ significantly. Future research should include a broader range of studies from diverse cultural and economic backgrounds to better understand how AI and VR can be adapted across various educational systems.

Another limitation is that the scope of this study is broad and includes multiple disabilities and technologies. As there are few studies that specifically address AI and VR in the context of disability, a general approach was taken to provide an initial overview. Although this provides a useful initial overview, an insight into specific technologies for each type of disability would likely provide better and more direct insights. For future research in this area, it is recommended to focus on illustrating more targeted practical applications by conducting studies with one disability and one technology at a time.

Additionally, the review predominantly features cross-sectional studies providing a snapshot of AI and VR's impact at a single point. Longitudinal studies tracking long-term effects are lacking but critical for understanding the sustained impact on learning outcomes. As Mielniczuk [72] argues, longitudinal studies are needed to evaluate the stability and long-term benefits or drawbacks of educational interventions. Future research should prioritise longitudinal studies exploring how students with disabilities engage with AI and VR over extended periods, offering insights into their lasting effects on both academic performance and social inclusion.

Furthermore, the search strategy may have inadvertently excluded relevant studies not explicitly using terms like "artificial intelligence" or "virtual reality". The focus on peer-reviewed journals indexed in Q1 and Q2 of the SCOPUS database may have also excluded valuable insights from books, conferences, or reports on emerging AI and VR trends. Future systematic reviews should expand search strategies to include a wider range of sources for a more comprehensive understanding of this rapidly evolving field.

In addition, this review focused specifically on AI and VR technologies, excluding extended technologies (XR) but including augmented reality (AR). This decision was based on the inclusion criteria, which prioritised studies published in high-impact, peer-reviewed Q1 and Q2 journals that provided empirical evidence of technologies demonstrating significant impact in supporting students with disabilities. While XR and AR have promising potential, especially in enhancing visual and contextual learning by overlaying digital content onto the physical world, their current application in fully immersive educational experiences for students with disabilities is relatively limited compared to AI and VR. This exclusion represents a limitation, as XR technologies could offer additional benefits not explored in this review. Future research should aim to include XR and AR technologies to provide a more comprehensive understanding of how they may complement AI and VR, offering new opportunities for inclusive education.

Moreover, the perspectives of teachers and educators, which play an important part in the successful implementation of AI and VR in education have not been comprehensively explored in this review. Further qualitative data collection from teachers and educators about the difficulties they experience and the type of assistance or training they would need to make such integration more effective is needed. There is an urgent need to explore these vectors in order to address practical constraints and create more flexible solutions regarding the incorporation of AI and VR in various educational contexts.

Finally, this review of AI and VR technologies has neglected to consider how other emerging technologies complement these technologies, such as Augmented Reality (AR) and machine learning (ML). Further studies should address how the effective use of different technologies, such as smart computers, synchronous communication devices and built environment features, could enhance the learning environment to become more interactive, inclusive and ultimately more useful for students with disabilities.

7. Practical Implications

The findings of this review have significant implications for both educational practice and theory. From a practical standpoint, integrating AI and VR technologies necessitates a shift in teaching strategies, emphasising personalised instruction and greater reliance on technology-mediated learning environments. Scholarly work by McGrath et al. [54] and Page et al. [21] underscores the crucial role of teacher training in ensuring the successful implementation of these technologies. Without adequate preparation and support, both teachers and educators may struggle to effectively incorporate AI and VR tools, potentially limiting their impact on students with disabilities.

Regarding policymakers, the financial and ethical challenges identified in this review must be addressed. Governments and educational institutions should explore public–private partnerships and other funding mechanisms to reduce the cost of AI and VR technologies, making them accessible to all schools, particularly those in under-resourced areas. Furthermore, ethical frameworks, such as those proposed by Brewer [7], are essential for guiding the responsible use of AI and VR in education, addressing concerns about data privacy, algorithmic bias, and equitable access. Establishing ethics committees within educational institutions can further ensure these technologies promote fairness and inclusivity. This includes focusing on ensuring equity not only within high-income regions but also facilitating access and adoption in lower-income areas where resources for such technologies may be limited.

Moreover, when companies develop AI systems to assist students with disabilities, they should also take into consideration that these are accessible and easy to use [73]. The principle of equal and fair treatment of all students involves understanding all students' needs to promote each student's participation in the learning process. Involving professionals such as psychologists, educators and doctors in the development of these systems can improve their effectiveness and take into account the different needs of students with disabilities. Educational leaders should actively engage in such partnerships, advocating for tools that are accessible, functional and inclusive. Additionally, ensuring that AI and VR technologies are designed with scalability in mind is preponderant, so these tools can be effectively deployed across various educational settings, particularly in low-resource schools.

In addition, policymakers and education leaders should launch pilot programmes to evaluate the effectiveness of AI and VR technologies before these are fully deployed, ensuring adaptation to the diversity of educational environments before widespread adoption.

Theoretically, this review supports constructivist and experiential learning theories by demonstrating how AI and VR can create more engaging and personalised learning experiences. Immersive VR environments, in particular, offer opportunities for hands-on, experiential learning that aligns with Dewey's theories of education as experience [74]. However, the review also highlights the need for a critical pedagogy that questions the power dynamics inherent in AI-driven educational tools and advocates for their responsible

use. As Friere [75] argues, technology in education must be used to empower students rather than reinforce existing hierarchies or inequalities. Future research should also explore how AI and VR can further support marginalised or disadvantaged groups in education to ensure that these technologies serve as tools for empowering diverse student populations.

8. Conclusions

The integration of AI and VR technologies into education offers significant potential for improving the inclusion of students with disabilities. These technologies offer personalised learning experiences, immersive environments and the ability to tailor educational content to individual needs, enabling a more inclusive and engaging learning experience.

Despite their promising possibilities, challenges such as high costs, technical limitations and insufficient teacher preparation need to be addressed in order to fully exploit the potential of AI and VR in education. Ethical considerations, including data privacy and algorithmic bias, require careful management to ensure equitable access and avoid exacerbating existing educational disparities.

The effective inclusion of AI and VR technologies requires comprehensive teacher/educator training and supportive institutional frameworks. Policymakers and educational institutions should consider funding mechanisms and partnerships to make these technologies accessible, especially in under-resourced settings. Establishing ethical guidelines is important to guide the responsible use of AI and VR and ensure that they empower students and enhance learning outcomes.

The use of AI and VR technologies is in line with the objectives of the CRPD, which promotes inclusive education and provides equitable learning opportunities for all students regardless of their background and abilities. By addressing the challenges identified and harnessing the potential of these new technologies, educational environments can become more inclusive and accessible, enabling students with disabilities to reach their full potential.

Author Contributions: Conceptualization, S.M., A.K. and A.C.; methodology, S.M.; software, S.M.; validation, S.M., A.C. and A.K.; formal analysis, A.C., A.S., M.K., A.M., M.M. and S.M.; investigation, S.M. and A.C.; resources, M.K., A.M. and M.M.; data curation, A.C. and S.M.; writing—original draft preparation, S.M., A.C. and A.S.; writing—review and editing, M.K., A.M. and M.M.; visualisation, M.K., A.S., A.M. and M.M.; supervision, S.M. and A.S.; project administration, S.M. and A.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not Applicable.

Data Availability Statement: No new data were created or analyzed in this study.

Conflicts of Interest: The authors declare no conflict of interest.

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